Homework 4: Concise Answers

CSP1 (p. 1)

The eight edges are

$$E = \{ab, ac, ad, bc, be, cd, ce, de\}, \qquad x_u \neq x_v \quad (uv \in E).$$

After applying the unary constraints:

$$D(a) = \{1, 2\}, \quad D(b) = \{1, 2, 3\}, \quad D(c) = D(d) = D(e) = \{1, 2, 3, 4\}.$$

Selection rules. Interpret *minimum value heuristic* as minimum remaining values (MRV): choose the unassigned variable with the smallest current domain. Break MRV ties by the largest number of unassigned neighbors (degree heuristic), then alphabetically. Try values in increasing order. After each assignment, use forward checking to remove its value from every unassigned neighbor's domain.

Step	Assignment	$D(a)$	$D(b)$	$D(c)$	$D(d)$	$D(e)$
0	–	$\{1,2\}$	$\{1,2,3\}$	$\{1,2,3,4\}$	$\{1,2,3,4\}$	$\{1,2,3,4\}$
1	$a=1$	–	$\{2,3\}$	$\{2,3,4\}$	$\{2,3,4\}$	$\{1,2,3,4\}$
2	$b=2$	–	–	$\{3,4\}$	$\{2,3,4\}$	$\{1,3,4\}$
3	$c=3$	–	–	–	$\{2,4\}$	$\{1,4\}$
4	$d=2$	–	–	–	–	$\{1,4\}$
5	$e=1$	–	–	–	–	–

A dash in a domain column means that the variable is already assigned.

Step 1. a has the smallest domain (size 2). Set $a = 1$; delete 1 from $D(b), D(c), D(d)$.

Step 2. b now has the smallest domain (size 2). Set $b = 2$; delete 2 from $D(c), D(e)$.

Step 3. c now has the smallest domain (size 2). Set $c = 3$; delete 3 from $D(d), D(e)$.

Step 4. d, e both have domain size 2 and one unassigned neighbor. The degree heuristic ties, so choose d alphabetically. Set $d = 2$. Since $2 \notin D(e)$, $D(e)$ does not change.

Step 5. Set $e = 1$. No domain becomes empty; no backtracking is needed.

$$\boxed{(a, b, c, d, e) = (1, 2, 3, 2, 1)}.$$

For edges ordered as $ab, ac, ad, bc, be, cd, ce, de$, the endpoint values are

$$((1,2), (1,3), (1,2), (2,3), (2,1), (3,2), (3,1), (2,1)).$$

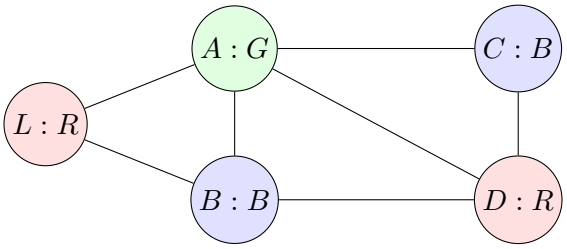
Every pair has different values, and both unary constraints hold.

CSP2 (p. 2)

Label the left node L, upper-left A, lower-left B, upper-right C, and lower-right D. The edges are $LA, LB, AB, AC, AD, BD, CD$.

Since $L = R$, the adjacent nodes A, B cannot be red. Also $A \neq B$, so choose $A = G,\ B = B$ (blue). Then D is adjacent to both A and B, hence $D = R$. Finally C is adjacent to A and D, hence $C = B$.

$$\boxed{(L, A, B, C, D) = (R, G, B, B, R)}.$$



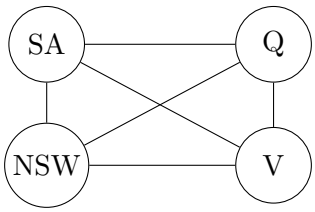
All seven edges join different colors.

CSP3 (p. 3)

Write SA = South Australia, Q = Queensland, NSW = New South Wales, V = Victoria, WA = Western Australia, NT = Northern Territory.

In the modified map, Victoria extends along the east side of NSW and shares a border with Queensland. Thus SA, Q, NSW, V are pairwise adjacent:

$$\{SA{-}Q,\ SA{-}NSW,\ SA{-}V,\ Q{-}NSW,\ Q{-}V,\ NSW{-}V\}.$$



These four regions form K_4. With only three colors, at least two would have the same color, contradicting their adjacency. Therefore $\chi \geq 4$.

Four colors also suffice. One valid coloring is

Region	WA	NT	SA	Q	NSW	V
Color	2	3	1	2	3	4

Every neighboring pair has different colors. Hence $\chi \leq 4$, and

$$\boxed{\chi = 4}.$$